AFFINENESS APPEARS AT A FINITE STAGE IN THE DESCENT SITUATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1854-1862, following the statement at 1848-1852.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Affineness appears at a finite stage. lemma-limit-is-affine (same chapter, card L26293, cited at ~1858-1859). V_i x_{Y_i} X_i is affine for some i.
- Affine is etale-local on the target. Morphisms of Spaces, Lemma spaces-morphisms-lemma-affine-local (cross-chapter, cited at ~1860-1861). f_i affine detected on the etale cover.

Source reference. Original lemma 07SQ: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1370–1370. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1666–1666. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03WG: focused-seeds/geometry/S9-spaces-limits-descend-affine/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Notation and assumptions as in Situation 084W. If f is affine, then f_i is affine for some $i \geq 0$.*

Proof. Choose an affine scheme V_0 and a surjective étale morphism $V_0 \to Y_0$. Set $V_i = V_0 \times_{Y_0} Y_i$ and $V = V_0 \times_{Y_0} Y$. Since f is affine we see that $V \times_Y X = \lim V_i \times_{Y_i} X_i$ is affine. By Lemma 07SQ we see that $V_i \times_{Y_i} X_i$ is affine for some $i \geq 0$. For this i the morphism f_i is affine (Morphisms of Spaces, Lemma 03WG). □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.